RESTAURANT GROWTH POTENTIAL MODELLING & STRATEGIC CLASSIFICATION SYSTEM

AN END-TO-END DATA-DRIVEN ANALYSIS FOR EXPANSION READINESS

Name: Hari Ganesh S A

Gmail: hariganeshsa@gmail.com (payment)

Gmail for Project: hs2110@srmist.edu.in

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1. Introduction

Restaurant expansion is one of the most capital-intensive and risk-prone strategic decisions in the food service industry. While traditional decision-making often relies on surface-level indicators such as revenue growth or order volume, such approaches frequently fail to account for **cost pressure, channel dependency, scalability constraints, and structural inefficiencies**. As a result, businesses often expand high-volume restaurants that later struggle to sustain profitability.

This project was designed to move beyond simplistic growth metrics and build a **holistic, data-driven framework** for evaluating restaurant expansion readiness. By integrating **exploratory data analysis (EDA), feature engineering, clustering algorithms, and a Growth Potential Index (GPI)**, the analysis provides actionable insights into which restaurants are truly positioned for sustainable expansion and which require structural optimization before scaling.

The study uses real operational, financial, and channel-level data to classify restaurants into strategic archetypes and identify the **key levers that drive or constrain expansion success**.

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2. Business Problem Statement

The central business problem addressed in this project is:

How can restaurants be objectively classified based on their true expansion readiness, rather than superficial growth indicators, while identifying the operational levers that most influence sustainable scaling?

Key challenges addressed:

- High order volume masking low profitability
- Heavy dependence on third-party aggregators
- Rising cost pressure during aggressive growth
- Lack of a unified metric to assess expansion readiness
- Absence of strategic segmentation for decision-making

To solve this, the project focuses on three core analytical pillars:

1. **Holistic Growth-Readiness Assessment**
2. **Data-Driven Restaurant Classification**
3. **Identification of Growth Levers Impacting Expansion Readiness**

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3. Data Understanding and Preparation

3.1 Dataset Overview

The dataset consists of restaurant-level operational and financial data, including:

- Orders across channels (In-store, Uber Eats, DoorDash, Self-Delivery)
- Revenue by channel
- Cost components (COGS, OPEX, commissions)
- Growth metrics (Growth Factor, Monthly Orders)
- Business attributes (Segment, Cuisine Type)
- Geographic attributes (Subregion)

The working dataset contained **301+ columns** after feature engineering, enabling multi-dimensional analysis.

3.2 Data Cleaning Process

The cleaning process focused on ensuring **business realism and analytical consistency**:

- Missing values in operational metrics were validated and corrected
- Outliers in order volumes and cost ratios were reviewed
- Channel-wise orders were cross-checked against total monthly orders
- Derived metrics were recalculated where inconsistencies were found

This ensured that all downstream analyses were based on **reliable and interpretable data**.

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4. Feature Engineering, Encoding, and Scaling

4.1 Feature Engineering

Key derived features included:

- Aggregator dependency ratios
- Channel-level contribution percentages
- Cost pressure indicators
- Net profit and margin proxies

- Growth Potential Index (GPI)

4.2 Encoding

Categorical variables such as:

- Segment
- Cuisine Type
- Subregion

were encoded using **One-Hot Encoding**, strictly for machine learning purposes.

4.3 Scaling

Numerical features were scaled using **StandardScaler** to ensure:

- No dominance by high-magnitude variables
- Fair distance calculations for clustering algorithms

Importantly, **scaled and encoded data were used only for modeling**, while **raw data was used for all business interpretation and visualization**.

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5. Growth Potential Index (GPI)

5.1 Purpose of GPI

The Growth Potential Index was developed to capture **expansion readiness** by combining:

- Demand strength
- Revenue quality
- Cost resilience
- Channel scalability

5.2 GPI Construction Logic

Positive contributors:

- Growth Factor
- Monthly Orders
- AOV
- Delivery Reach

Negative contributors:

- COGS Rate
- OPEX Rate
- Commission Rate

Higher GPI values indicate **greater readiness for scalable expansion**, not merely higher current performance.

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6. Holistic Growth-Readiness Assessment

This analysis evaluates growth readiness at the **portfolio, geographic, segment, and restaurant level**, using multiple EDA charts.

Key Findings and Interpretation

1️⃣ **Stable Performers dominate the portfolio (44.4%)**, yet generate only **~52% of High-Growth monthly orders**.
This indicates that demand is **concentrated in a smaller subset of restaurants**, increasing dependency risk on a limited number of outlets.

2️⃣ **High-Growth restaurants (19.0%) produce the highest average monthly orders (~1,531)** but show a **negative average GPI (-0.08)**.
This reveals that growth in this cluster is driven by **cost-heavy expansion**, where demand increases faster than profitability.

3️⃣ **Aggregator-Dependent Low-Margin restaurants (36.6%) exhibit the highest average GPI (+0.31)**.
Despite heavy reliance on aggregators, these restaurants demonstrate **efficient growth mechanics**, proving that platform dependence does not automatically negate growth potential.

4️⃣ **South Auckland leads geographically with the highest average GPI (+0.34)**, while the **CBD records the lowest GPI (-0.58)** despite comparable order volumes.
This highlights that **cost structures and regional economics** matter more than raw demand when evaluating expansion.

5️⃣ **QSRs represent the largest segment (32%)**, but **Ghost Kitchens (11.6%) deliver the highest average GPI (+0.12)**.
This demonstrates that **scalability and asset-light models** outperform size-driven models in growth readiness.

6️⃣ **The top 10 restaurants alone generate ~1,550–1,770 monthly orders each**, confirming that growth is **highly skewed toward a small elite group**.
This reinforces the need for **targeted expansion strategies**, rather than broad-based scaling.

Solution & Recommendations (Holistic Growth-Readiness)

- Do not equate volume with readiness
- Prioritize high-GPI regions and business models
- Reduce cost pressure before scaling High-Growth outlets
- Diversify demand away from over-concentrated performers

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7. Data-Driven Restaurant Classification Analysis

This analysis classifies restaurants using **orders, revenue, cuisine behavior, and GPI**, moving beyond traditional segmentation.

Key Findings and Interpretation

1️⃣ **Aggregator dominance is structurally high**
Uber Eats contributes the **largest share of total orders (~800K)** and the **highest revenue (~18K)**, confirming strong demand but also **systemic platform dependency risk**.

2️⃣ **Revenue does not scale proportionally with orders across channels**
Self-Delivery and DoorDash generate **comparable revenue with significantly fewer orders** than Uber Eats, indicating **superior unit economics per order**.

3️⃣ **High-volume cuisines are not always high-growth cuisines**
Pizza and Burgers lead in monthly orders, but **only Pizza shows positive GPI**, proving that demand alone does not imply expansion readiness.

4️⃣ **Cuisine-level monetization differs sharply from demand**
Burgers and Indian cuisine generate high order volumes, yet their **GPI is weak or negative**, signaling cost pressure or channel imbalance.

5️⃣ **Only a few cuisines are growth-ready**
Pizza and Burgers are the **only cuisines with positive GPI**, while Thai, Kebabs/Mediterranean, and Indian cuisines exhibit **negative GPI**, marking expansion risk.

6️⃣ **Strategic implication**
Growth strategies should **prioritize high-GPI cuisines** while **restructuring cost and channel mix** for high-volume but low-GPI cuisines.

Solution & Recommendations (Classification Analysis)

- Treat cuisine as a strategic lever, not just a branding choice
- Rebalance channels for low-GPI cuisines
- Avoid expanding cuisines with structurally negative GPI
- Use classification to guide localized strategy decisions

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8. Growth Levers Impacting Expansion Readiness

This analysis focuses on identifying **which operational levers truly matter for expansion**.

Key Findings and Interpretation

1️⃣ **High-Growth restaurants process ~55% of orders via aggregators**, the lowest dependency among clusters.
This indicates **better channel control**, a critical factor for scalable growth.

2️⃣ **High-Growth cluster leads demand with ~1,531 monthly orders**, nearly **90% higher than Stable Performers (804)**.
Demand remains the **strongest positive expansion signal**.

3️⃣ **Average Order Value (₹38–₹39) is flat across all clusters**, contributing **less than 3% variation**.
This proves that **AOV is not a differentiating growth lever** in the current dataset.

4️⃣ **Cost pressure peaks in High-Growth restaurants (COGS + OPEX ≈ 88%)**.
Despite strong demand, **cost inefficiency is the primary expansion risk**, making cost control the most critical lever.

Solution & Recommendations (Growth Levers)

- Focus on channel mix optimization
- Treat demand as necessary but not sufficient
- Stop relying on AOV improvements alone
- Enforce cost ceilings before expansion approval

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9. Machine Learning Models & Interpretation

9.1 K-Means Clustering (PCA Visualization)

What the model shows:

- Clear separation into three dense clusters
- Partial overlap, expected in real business data
- PCA used strictly for visualization

Cluster Interpretations

🟣 **High-Growth / High-Risk**

- High orders and growth signals
- Wide delivery reach
- High volatility and cost pressure
  👉 *Strategic meaning:* Scale cautiously with strict cost controls

🟢 **Stable Local Performers**

- Balanced orders and AOV
- Controlled costs
- Predictable performance
  👉 *Strategic meaning:* Optimize, not aggressively expand

🟡 **Aggregator-Dependent Low Margin**

- Heavy platform reliance
- High commissions
- Volume without margin
  👉 *Strategic meaning:* Rebalance channels before scaling

9.2 Hierarchical Clustering

- Clear top-level split into **three branches**
- Sub-clusters exist within each group
- Confirms multi-level similarity structure

Business conclusion:
Restaurants naturally organize into **three strategic segments**, validating K-Means as the deployment model.

9.3 DBSCAN

- Most points labeled as noise
- No dense clusters detected

Conclusion:
❌ Not suitable for segmentation
✅ Used only as a robustness check

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10. Final Strategic Recommendations (PPT – 10 Points)

1. Scale only high-GPI restaurants
2. Reduce aggregator dependency below 55%
3. Enforce cost ceilings before expansion
4. Prioritize South Auckland and similar regions
5. Avoid CBD-style cost-heavy expansion
6. Invest in Ghost Kitchen scalability
7. Rebalance channels for low-margin cuisines
8. Diversify growth away from top-10 concentration
9. Use clustering for quarterly portfolio reviews
10. Treat GPI as a gating metric for expansion approval

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11. Conclusion

This project demonstrates that **sustainable restaurant expansion is not driven by demand alone**, but by a complex interaction of **channel control, cost efficiency, regional economics, and business model design**. By combining advanced EDA, machine learning, and strategic scoring, the framework provides a **robust, repeatable, and executive-ready decision system**.

This project shows that **sustainable restaurant growth is driven by structural efficiency, not just high demand**. By combining EDA, a Growth Potential Index (GPI), and machine learning–based classification, the analysis reveals that many top GPI restaurants are not in the High-Growth cluster, proving that cost control, channel balance, and business model design matter more than raw order volume. High-Growth restaurants generate strong demand but face heavy cost pressure, while some aggregator-dependent restaurants achieve higher growth readiness through operational efficiency.

Going forward, expansion decisions should be based on **GPI rather than volume**, focusing on cost-efficient regions and scalable business models. Restaurants should not be expanded solely on demand spikes or aggregator-driven growth without margin safeguards. This framework provides a data-driven approach to decide **where to scale, where to optimize, and where to restructure before expansion**.

Final Question:

"Which restaurants are structurally positioned for sustainable growth, and how should they be classified?

Answer:
Restaurants that are structurally positioned for sustainable growth are those that combine strong and consistent demand, controlled cost structures, balanced channel mix (lower aggregator dependency), and positive Growth Potential Index (GPI) scores.

Based on these structural characteristics, restaurants should be classified into three strategic categories:

1 High-Growth / High-Risk
*Restaurants with high monthly orders and strong demand momentum but elevated cost pressure. They are positioned for growth **only if costs and channel dependency are actively managed**.*

2 Stable Local Performers
*Restaurants with balanced demand, controlled costs, and predictable performance. They represent **sustainable but incremental growth opportunities**, suitable for optimization rather than aggressive expansion.*

3 Aggregator-Dependent Low-Margin
*Restaurants that generate volume primarily through third-party platforms but suffer from high commission and limited margin control. These units are **structurally constrained** and require channel rebalancing before any expansion.*